

- 1.4 g.  $\therefore$  the pressure, volume, and temperature determine the number of moles present from the ideal gas law, multiplying that amount by the molar mass gives the sample mass.
- $1.29 \times 10^4$  Pa.  $\therefore$  the balanced equation determines the amount of  $\text{H}_2\text{O}$  produced from the amount of  $\text{NaHCO}_3$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
- $3.30 \times 10^4$  Pa  $\therefore$  Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
- 3.73 L.  $\therefore$  the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product  $PV$  remains unchanged.
- 0.43. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
- 0.15 m.  $\therefore$  Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
- 0.48.  $\therefore$  23 g of hydrogen peroxide corresponds to 0.68 mol. The reaction produces 0.34 mol of oxygen. The amount of argon remains 0.318 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
- $5.4 \times 10^3$  Pa.  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
- $6.9 \times 10^3$  Pa.  $\therefore$  the balanced equation determines the amount of  $\text{H}_2$  produced from the amount of  $\text{HCl}$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
- HBr.  $\therefore$  Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 80.91 g/mol. Only HBr has that molar mass, so it is the only identity consistent with the measurements.
- Increases.  $\therefore$  increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.
- $9.9 \times 10^4$  Pa  $\therefore$  the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.
- $2.02 \times 10^4$  Pa.  $\therefore$  the balanced equation determines the amount of  $\text{C}_2\text{H}_2$  produced from the amount of  $\text{CaC}_2$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
- $13 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:2 ratio of  $\text{N}_2\text{O}_4$  to  $\text{NO}_2$ , so the product volume is multiplied by 2/1.
- $1.33 \times 10^3$  mol.  $\therefore$  the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
- 110 kPa.  $\therefore$  the gas sample and its temperature do not change,  $PV$  is constant. The smaller final volume produces a proportionally larger pressure, so the initial pressure must be lower by the inverse volume ratio.
- No.  $\therefore$  keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
- 0.11 m.  $\therefore$  Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
- 2.44 L.  $\therefore$  the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
- 1.7 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
- $7.18 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The

balanced equation shows a 1:2 particle ratio between  $\text{C}_2\text{H}_5\text{OH}$  and  $\text{CO}$ , so their volumes must be in the same ratio.

22.  $5.73 \times 10^3 \text{ Pa}$ .  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
23.  $37 \text{ g/mol}$ .  $\therefore$  density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.
24.  $9.39 \times 10^4 \text{ Pa}$ .  $\therefore$  the measured pressure is the sum of the partial pressures of hydrogen and water vapor, so the water vapor pressure must be subtracted to obtain the hydrogen pressure before converting to SI units.
25.  $5.12 \text{ mL}$ .  $\therefore$  at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 2:2 mole ratio, so the gas volumes must change in the same 2:2 ratio.
26.  $9.12 \text{ L}$ .  $\therefore$  the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
27.  $0.35 \text{ atm}$ .  $\therefore$  the amount of the second gas is the difference between the total amount and the known amount, and each gas produces pressure according to its own amount through  $P = nRT/V$ .
28.  $147 \text{ N}$  The piston is square, so its area is  $\text{side}^2$ .  $\therefore$  pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
29.  $5.05 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 1:1 particle ratio between  $\text{Cl}_2$  and  $\text{H}_2$ , so their volumes must be in the same ratio.
30.  $1.6 \times 10^4 \text{ Pa}$ .  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
31.  $27.7 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:7 particle ratio between  $\text{C}_2\text{H}_6$  and  $\text{O}_2$ , so their volumes must be in the same ratio.
32.  $1.0 \text{ g}$ .  $\therefore$  the pressure, volume, and temperature determine the number of moles present from the ideal gas law, multiplying that amount by the molar mass gives the sample mass.
33.  $0.875 \text{ mL}$ .  $\therefore$  at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for  $\text{H}_2:\text{C}_2\text{H}_5\text{OH}$  is 4:1 so the volume changes by the same factor.
34.  $6.54 \times 10^4 \text{ Pa}$ .  $\therefore$  Dalton's law states that the total pressure of a gas mixture equals the sum of the partial pressures because each gas contributes independently to the total pressure.
35. 135% of the initial volume.  $\therefore$  rearranging the combined gas law gives  $V_2/V_1 = (T_2/T_1)(P_1/P_2)$ . The final volume is determined by the competing effects of the change in absolute temperature and the change in pressure.
36.  $0.13 \text{ mol}$ .  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
37.  $1.82 \times 10^4 \text{ Pa}$ .  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
38.  $1.0 \times 10^5 \text{ Pa}$ .  $\therefore$  water evaporates until its vapor reaches its equilibrium vapor pressure, so the collected sample contains both gases and their partial pressures add to give the total pressure.
39.  $1.60 \times 10^5 \text{ Pa}$ .  $\therefore$  Dalton's law gives  $P_{\text{gas}} = x_{\text{gas}}P_{\text{total}}$ , so dividing the partial pressure by the mole fraction gives the total pressure before converting to SI units.
40.  $1.09 \times 10^3 \text{ mol}$ .  $\therefore$  the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
41.  $9.13 \times 10^4 \text{ Pa}$ .  $\therefore$  the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.
42.  $20. \text{ cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the  $\text{O}_2:\text{H}_2\text{O}$  coefficient ratio of 2:1 determines the required volume.
43.  $1.4 \text{ g}$ .  $\therefore$  the pressure, volume, and temperature determine the number of moles present from the ideal gas law, multiplying that amount by the molar mass gives the sample mass.
44.  $0.055 \text{ mol}$ .  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured

pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .

45. 5.6 L.  $\therefore$  the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product  $PV$  remains unchanged.
46.  $6.28 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 1:2 ratio of  $\text{Cl}_2$  to  $\text{HCl}$ , so the product volume is multiplied by 2/1.
47. Yes; final temperature =  $149.^\circ\text{C}$ .  $\therefore$  the amount of gas and volume remain constant, the ideal gas law requires pressure and absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.
48. Decreases.  $\therefore$  lowering the temperature reduces the kinetic energy of the liquid molecules, so fewer molecules have sufficient energy to enter the vapor phase and the equilibrium vapor pressure decreases.
49. 0.571 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
50. 11.3 L.  $\therefore$  the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
51. 190 N The piston is square, so its area is  $\text{side}^2$ .  $\therefore$  pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
52.  $52 \text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:8 ratio of  $\text{C}_4\text{H}_{10}$  to  $\text{CO}_2$ , so the product volume is multiplied by 8/2.
53.  $5.69 \text{ cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the  $\text{SO}_2$ : $\text{SO}_3$  coefficient ratio of 1:1 determines the required volume.
54. 2.6 mL.  $\therefore$  at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for  $\text{HCl}$ : $\text{Cl}_2$  is 2:1 so the volume changes by the same factor.
55. 216 K.  $\therefore$  rearranging the combined gas law gives  $T_2 = T_1(P_2V_2)/(P_1V_1)$ . Absolute temperature is directly proportional to the product of pressure and volume for a fixed amount of gas, so kelvin must be used.
56.  $1.43 \times 10^4 \text{ Pa}$ .  $\therefore$  the balanced equation determines the amount of  $\text{O}_2$  produced from the amount of  $\text{H}_2\text{O}_2$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
57. Remains unchanged.  $\therefore$  the container volume is fixed and the amount of gas remains constant, both the mass and the volume stay constant. Heating changes the pressure, not the mass per unit volume.
58. Consistent.  $\therefore$  The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which matches the molar mass of  $\text{H}_2\text{S}$ .
59. 0.66.  $\therefore$  8.7 g of ammonium chloride corresponds to 0.16 mol. The reaction produces 0.16 mol of ammonia. The amount of helium remains 0.32 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
60. 0.127 m.  $\therefore$  Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
61. Not consistent.  $\therefore$  Combining the measured density with the ideal gas law gives the experimental molar mass, which does not match the molar mass of  $\text{C}_2\text{H}_6$ .
62. Not consistent.  $\therefore$  The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which does not match the molar mass of He.
63.  $8.15 \times 10^4 \text{ Pa}$   $\therefore$  Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
64. 2.05 L.  $\therefore$  the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
65. 649 mol.  $\therefore$  the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .

66. 42 g/mol.  $\therefore$  the measured pressure, volume, and temperature determine the number of moles through the ideal gas law, dividing the measured mass by that amount gives the molar mass.
67. 13 cm<sup>3</sup>.  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:8 ratio of C<sub>4</sub>H<sub>10</sub> to CO<sub>2</sub>, so the product volume is multiplied by 8/2.
68. 6.8 mL.  $\therefore$  at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:1 mole ratio, so the gas volumes must change in the same 1:1 ratio.
69. Not consistent.  $\therefore$  Combining the measured density with the ideal gas law gives the experimental molar mass, which does not match the molar mass of N<sub>2</sub>O.
70. 0.109 m.  $\therefore$  Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
71. 0.20. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
72. Increases.  $\therefore$  increasing the temperature increases the average kinetic energy of the liquid molecules, allowing more molecules to escape into the vapor phase before dynamic equilibrium is reestablished, producing a higher equilibrium vapor pressure.
73. 0.466 atm; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
74. 3.38 cm<sup>3</sup>.  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:1 ratio of SO<sub>3</sub> to O<sub>2</sub>, so the product volume is multiplied by 1/2.
75.  $7.7 \times 10^4$  Pa  $\therefore$  the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.
76. 6.85 mL.  $\therefore$  at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 2:1 mole ratio, so the gas volumes must change in the same 2:1 ratio.
77. 2.5 L.  $\therefore$  the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
78. 2.94 cm<sup>3</sup>.  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 2:1 particle ratio between HBr and Br<sub>2</sub>, so their volumes must be in the same ratio.
79. 27 N The piston is square, so its area is side<sup>2</sup>.  $\therefore$  pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
80. 30% of the initial volume.  $\therefore$  rearranging the combined gas law gives  $V_2/V_1 = (T_2/T_1)(P_1/P_2)$ . The final volume is determined by the competing effects of the change in absolute temperature and the change in pressure.
81. 150 N The piston is square, so its area is side<sup>2</sup>.  $\therefore$  pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
82.  $7.6 \times 10^3$  Pa.  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
83. 3.4 L.  $\therefore$  the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product  $PV$  remains unchanged.
84. 4.7 cm<sup>3</sup>.  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 1:1 particle ratio between H<sub>2</sub>O and CO<sub>2</sub>, so their volumes must be in the same ratio.
85. 88 g/mol.  $\therefore$  the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
86. 647 mol.  $\therefore$  the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
87. Xe.  $\therefore$  Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 131.29 g/mol. Only Xe has that molar mass, so it is the only identity consistent with the measurements.
88. Not consistent.  $\therefore$  The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which does not match the molar mass of Ne.
89.  $9.6 \times 10^4$  Pa  $\therefore$  the measured pressure is the sum of the partial pressures of the dry gas and water vapor. Subtracting the water

vapor pressure isolates the pressure exerted by the dry gas before converting to SI units.

90.  $1.02 \times 10^5 \text{ Pa}$  ∴ the total pressure equals the sum of the partial pressures of all gases present, including the water vapor, before converting to SI units.
91. Consistent. ∴ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which matches the molar mass of HF.
92.  $6.1 \times 10^3 \text{ Pa}$  ∴ a closed-end manometer has a vacuum at one end, so the height difference equals the absolute gas pressure before converting to SI units.
93.  $50. \text{ g/mol}$ . ∴ density gives the mass per unit volume, while the ideal gas law relates volume to the number of moles, combining these relationships yields the mass of one mole of the gas.
94. Not consistent. ∴ The measured pressure, volume, and temperature determine the amount of gas using the ideal gas law. Dividing the measured mass by that amount gives the experimental molar mass, which does not match the molar mass of  $\text{C}_3\text{H}_8$ .
95. NO. ∴ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately  $30.01 \text{ g/mol}$ . Only NO has that molar mass, so it is the only identity consistent with the measurements.
96.  $36 \text{ g/mol}$ . ∴ density gives the mass per unit volume, while the ideal gas law relates volume to the number of moles, combining these relationships yields the mass of one mole of the gas.
97.  $13.7 \text{ mL}$ . ∴ at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 2:2 mole ratio, so the gas volumes must change in the same 2:2 ratio.
98.  $4.53 \text{ L}$ . ∴ the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
99.  $1.8 \text{ L}$ . ∴ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
100.  $3.7 \text{ L}$ . ∴ constant temperature and a fixed amount of gas require  $PV$  to remain constant. Because the final pressure is greater than the initial pressure, the initial volume must have been proportionally larger.
101.  $1.3 \times 10^4 \text{ Pa}$ . ∴ the balanced equation determines the amount of  $\text{NH}_3$  produced from the amount of  $\text{NH}_4\text{Cl}$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
102.  $4.0 \text{ cm}^3$ . ∴ at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 1:1 particle ratio between  $\text{CH}_4$  and  $\text{CO}_2$ , so their volumes must be in the same ratio.
103.  $0.272 \text{ atm}$ ; Equal to the external pressure. Boiling occurs only when the vapor pressure matches the surrounding pressure. If the vapor pressure were lower, bubbles would collapse, and if it were higher, boiling would not be at equilibrium.
104.  $0.73$ . ∴ removing one gas leaves only the pressure produced by the remaining gas, and pressure is proportional to the number of moles at constant temperature and volume.
105.  $1.0$ . ∴ both gases occupy the same volume at the same temperature, each pressure is directly proportional to its amount, so the pressure ratio equals the mole ratio.
106.  $\text{C}_2\text{H}_2$ . ∴ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately  $26.04 \text{ g/mol}$ . Only  $\text{C}_2\text{H}_2$  has that molar mass, so it is the only identity consistent with the measurements.
107.  $5.492 \times 10^4 \text{ Pa}$  ∴ Dalton's law states that the total pressure of a gas mixture equals the sum of the partial pressures because each gas contributes independently to the total pressure.
108.  $\text{H}_2$ . ∴ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately  $2.02 \text{ g/mol}$ . Only  $\text{H}_2$  has that molar mass, so it is the only identity consistent with the measurements.
109.  $0.75$ . ∴  $14 \text{ g}$  of potassium bicarbonate corresponds to  $0.14 \text{ mol}$ . The reaction produces  $0.14 \text{ mol}$  of carbon dioxide. The amount of neon remains  $0.43 \text{ mol}$ . The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
110. Remains unchanged. ∴ the container volume is fixed and the amount of gas remains constant, both the mass and the volume stay constant. Heating changes the pressure, not the mass per unit volume.
111.  $0.17$ . Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
112.  $41.4 \text{ N}$  The piston is square, so its area is  $\text{side}^2$ . ∴ pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
113.  $13 \text{ g/mol}$ . ∴ density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.

114. 24.3 L.  $\therefore$  the ideal gas law gives  $PV = nRT$ , but the amount of gas is not given directly. The amount must first be determined from the sample mass using  $n = m/M$ , after which solving  $PV = nRT$  for volume yields the correct result.
115.  $4.96\text{ cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the  $\text{HCl}:\text{NH}_3$  coefficient ratio of 0.5:1 determines the required volume.
116. 2.9 L.  $\therefore$  the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
117. Remains unchanged.  $\therefore$  the equilibrium vapor pressure of a pure substance depends only on temperature. Adding more liquid changes the amount of liquid present but does not change the equilibrium established between evaporation and condensation.
118. 3.0 L.  $\therefore$  the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
119.  $9.6 \times 10^4\text{ Pa}$   $\therefore$  the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
120. 1.11 atm; Yes. A liquid boils only when its vapor pressure equals or exceeds the external pressure, allowing vapor bubbles to remain stable throughout the liquid. Since the vapor pressure is greater than or equal to the atmospheric pressure, vapor bubbles can continue to grow.
121. Decreases.  $\therefore$  the molar concentration is unchanged at the same temperature and pressure, but each mole has less mass. Less mass therefore occupies the same volume.
122. No.  $\therefore$  keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
123. 0.34 atm.  $\therefore$  each gas contributes the same fraction of the total pressure as its fraction of the total number of moles, the mole ratio determines the pressure ratio.
124. 2.2 L.  $\therefore$  the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
125.  $11.1\text{ cm}^3$ .  $\therefore$  at the same temperature and pressure, gas volume is directly proportional to the number of gas particles. The balanced equation shows a 1:2 particle ratio between  $\text{Cl}_2$  and  $\text{HCl}$ , so their volumes must be in the same ratio.
126.  $1.15 \times 10^5\text{ Pa}$   $\therefore$  the mercury is higher on the open end, gas pressure exceeds atmospheric pressure, and the pressure difference is added to the atmospheric pressure before converting to SI units.
127. 5.9 L.  $\therefore$  the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
128. No.  $\therefore$  keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
129. Increases.  $\therefore$  compression reduces the gas volume without changing its mass. The same mass occupying a smaller volume results in a greater density.
130.  $1.0 \times 10^5\text{ Pa}$   $\therefore$  water evaporates until its vapor reaches its equilibrium vapor pressure, so the collected sample contains both gases and their partial pressures add to give the total pressure.
131. 31 g/mol.  $\therefore$  density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.
132. 1.4 g.  $\therefore$  the pressure, volume, and temperature determine the number of moles present from the ideal gas law, multiplying that amount by the molar mass gives the sample mass.
133.  $6.46\text{ cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the  $\text{N}_2\text{O}_4:\text{NO}_2$  coefficient ratio of 2:1 determines the required volume.
134. 3.32 mL.  $\therefore$  at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for  $\text{CaO}:\text{CO}_2$  is 1:1 so the volume changes by the same factor.
135. Yes; final temperature =  $667^\circ\text{C}$ .  $\therefore$  the amount of gas and volume remain constant, the ideal gas law requires pressure and

absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.

136. 6.78 mL.  $\therefore$  at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 1:2 mole ratio, so the gas volumes must change in the same 1:2 ratio.
137. 0.17 mol.  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
138. Increases.  $\therefore$  adding gas increases the mass while the container volume does not change. More mass per unit volume produces a greater density.
139. 0.58. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
140. 9.6 mL.  $\therefore$  at the same temperature and pressure, Avogadro's law states that gas volume is directly proportional to the amount of gas. The balanced equation gives a 4:5 mole ratio, so the gas volumes must change in the same 4:5 ratio.
141.  $2.2\text{ cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the HCl:H<sub>2</sub> coefficient ratio of 0.5:1 determines the required volume.
142. 2.9 L.  $\therefore$  the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
143. 356 K.  $\therefore$  rearranging the combined gas law gives  $T_2 = T_1(P_2V_2)/(P_1V_1)$ . Absolute temperature is directly proportional to the product of pressure and volume for a fixed amount of gas, so kelvin must be used.
144. 990. mol.  $\therefore$  the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
145. 4.7 L.  $\therefore$  the amount of gas and the temperature remain constant, pressure and volume are inversely proportional. Increasing the pressure requires the volume to decrease so that the product  $PV$  remains unchanged.
146.  $2.73 \times 10^5\text{ Pa}$ .  $\therefore$  the amount of gas is constant, so  $PV/T$  remains constant. Increasing the absolute temperature increases pressure, whereas increasing volume decreases pressure; both effects combine according to  $P_1V_1/T_1 = P_2V_2/T_2$ , and temperatures must be expressed in kelvin because the proportionality is with absolute temperature.
147. 5.34 mL.  $\therefore$  at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for CaO:CO<sub>2</sub> is 1:1 so the volume changes by the same factor.
148. 0.907 atm; No. A liquid boils only when its vapor pressure equals or exceeds the external pressure, allowing vapor bubbles to remain stable throughout the liquid. Since the vapor pressure is less than the atmospheric pressure, vapor bubbles cannot continue to grow.
149. 0.17 m.  $\therefore$  Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
150. Yes; final temperature = 148.°C.  $\therefore$  the amount of gas and volume remain constant, the ideal gas law requires pressure and absolute temperature to change by the same factor. The required final temperature remains above the boiling point, so the substance can remain gaseous while the volume is held constant.
151. 0.17 mol.  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
152. 24 g/mol.  $\therefore$  density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.
153.  $1.5 \times 10^4\text{ Pa}$ .  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
154.  $1.1 \times 10^4\text{ Pa}$ .  $\therefore$  pressure is the applied force distributed over the piston face, only the piston cross-sectional area determines the pressure for a given force. The cylinder height does not affect the pressure produced by that applied force.
155.  $9.6 \times 10^4\text{ Pa}$ .  $\therefore$  the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.

156.  $9.2 \times 10^4 \text{ Pa}$  ∴ the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.
157.  $9.25 \times 10^4 \text{ Pa}$  ∴ the mercury is higher on the gas-sample side, the gas pressure is lower than atmospheric pressure, and the pressure difference is subtracted from the atmospheric pressure before converting to SI units.
158. Remains unchanged. ∴ the equilibrium vapor pressure of a pure substance depends only on temperature. Adding more liquid changes the amount of liquid present but does not change the equilibrium established between evaporation and condensation.
159. 8.2 L. ∴ the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
160.  $7.23 \times 10^4 \text{ Pa}$  ∴ the mole fractions must sum to 1, so the mole fraction of  $\text{N}_2$  is the remainder. Dalton's law states that a gas's partial pressure equals its mole fraction multiplied by the total pressure.
161. No. ∴ keeping the volume constant requires the absolute temperature to change in the same proportion as the pressure. The required final temperature would be below the boiling point, so condensation would occur and the sample could no longer remain entirely gaseous under the stated conditions.
162.  $3.76 \times 10^4 \text{ Pa}$  ∴ Dalton's law states that the partial pressure of a gas equals its mole fraction multiplied by the total pressure because each gas contributes independently to the total pressure.
163.  $1.10 \times 10^3 \text{ mol}$ . ∴ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
164. Remains unchanged. ∴ the equilibrium vapor pressure of a pure substance depends only on temperature. Adding more liquid changes the amount of liquid present but does not change the equilibrium established between evaporation and condensation.
165. Remains unchanged. ∴ the equilibrium vapor pressure of a pure substance depends only on temperature. Adding more liquid changes the amount of liquid present but does not change the equilibrium established between evaporation and condensation.
166.  $9.67 \times 10^4 \text{ Pa}$  ∴ the measured pressure includes both oxygen and water vapor. Subtracting the water vapor pressure gives the oxygen partial pressure before converting to SI units.
167. 20. g/mol. ∴ density gives the mass occupying a known volume, while the ideal gas law determines the amount of gas occupying that same volume. The mass per mole is therefore obtained by combining these relationships.
168. 0.148 m. ∴ Pressure is force distributed over an area, so the maximum allowable force fixes the largest permissible cross-sectional area through  $A = F/P$ . Because the opening is circular, the required geometric relationship between area and diameter determines the maximum allowable diameter.
169. 0.566. Mole fraction depends only on the relative numbers of moles present. No gas-law calculation is required because pressure, temperature, and volume do not affect the ratio of moles already present.
170. 0.75. ∴ 22 g of sodium nitrite corresponds to 0.32 mol. The reaction produces 0.16 mol of oxygen. The amount of helium remains 0.47 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
171. 806 mol. ∴ the container volume is fixed and the gas is at a specified pressure and temperature, the amount of gas is determined by the ideal gas law,  $PV = nRT$ . A higher pressure at the same volume and temperature corresponds to more gas particles being present, so the number of moles is found by rearranging the equation for  $n$ .
172. 0.057 mol. ∴ at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
173. 2.4 L. ∴ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.
174. Kr. ∴ Using the measured mass, pressure, volume and temperature, the sample's molar mass is approximately 83.80 g/mol. Only Kr has that molar mass, so it is the only identity consistent with the measurements.
175.  $1.65 \times 10^5 \text{ Pa}$  ∴ Dalton's law gives  $P_{\text{gas}} = x_{\text{gas}} P_{\text{total}}$ , so dividing the partial pressure by the mole fraction gives the total pressure before converting to SI units.
176. Increases. ∴ at the same temperature and pressure every ideal gas has the same molar concentration, but each mole of the replacement gas has a larger mass. More mass therefore occupies the same volume.
177. 3.5 L. ∴ the balloon is flexible, its internal pressure remains essentially equal to the surrounding atmospheric pressure. With the amount of gas unchanged, the ideal gas law gives volume directly proportional to absolute temperature, so the volume changes by the same ratio as the Kelvin temperatures.



178. 5.8L.  $\therefore$  the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
179.  $6.34\text{cm}^3$ .  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:2 ratio of NO to  $\text{NO}_2$ , so the product volume is multiplied by 2/2.
180. 26.6mL.  $\therefore$  at constant temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the corresponding volume ratio. The stoichiometric ratio for  $\text{O}_2:\text{H}_2\text{O}$  is 1:2 so the volume changes by the same factor.
181.  $2.38\text{cm}^3$ .  $\therefore$  at the same temperature and pressure, equal volumes of gases contain equal numbers of molecules (equivalently, equal moles), the gas-volume ratio is identical to the balanced-equation mole ratio. The reaction requires a 2:1 ratio of NOCl to  $\text{Cl}_2$ , so the product volume is multiplied by 1/2.
182. 0.301 mol.  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
183. 3.2L.  $\therefore$  the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
184. 0.44.  $\therefore$  both gases occupy the same volume at the same temperature, each pressure is directly proportional to its amount, so the pressure ratio equals the mole ratio.
185. 0.44. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
186.  $2.1 \times 10^4\text{Pa}$ .  $\therefore$  the balanced equation determines the amount of  $\text{O}_2$  produced from the amount of  $\text{H}_2\text{O}_2$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
187. 16g/mol.  $\therefore$  the ideal gas law determines the amount of gas from the measured pressure, volume, and temperature. Dividing the measured mass by that calculated amount gives the molar mass.
188.  $23.4\text{cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the  $\text{O}_2:\text{SO}_2$  coefficient ratio of 2:1 determines the required volume.
189. 0.83.  $\therefore$  18.7 g of barium carbonate corresponds to 0.0948 mol. The reaction produces 0.095 mol of carbon dioxide. The amount of neon remains 0.45 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.
190. 710kPa.  $\therefore$  the amount of gas and the temperature remain constant, the product  $PV$  remains constant. Reducing the volume causes molecules to collide with the container walls more frequently, increasing the pressure by the same factor that the volume decreases.
191.  $1.65\text{cm}^3$ . At the same temperature and pressure, gas volume is directly proportional to the amount of gas, so the balanced equation gives the same ratio between gas volumes as between stoichiometric coefficients. Therefore the NO: $\text{O}_2$  coefficient ratio of 0.5:1 determines the required volume.
192.  $9.5 \times 10^4\text{Pa}$ .  $\therefore$  the total pressure equals the sum of the partial pressures of all gases present, including the water vapor, before converting to SI units.
193. Remains unchanged.  $\therefore$  the container volume is fixed and the amount of gas remains constant, both the mass and the volume stay constant. Heating changes the pressure, not the mass per unit volume.
194.  $1.8 \times 10^3\text{Pa}$ .  $\therefore$  the balanced equation determines the amount of  $\text{O}_2$  produced from the amount of  $\text{Ag}_2\text{O}$  that reacts. Those moles of gas occupy the entire vessel, so the pressure is determined by the ideal gas law using the calculated amount of gas, the vessel volume, and the absolute temperature.
195. 0.50. Mole fraction depends only on the relative numbers of moles present. The given masses must first be converted to moles using the molar masses before calculating the ratio.
196. 0.17 mol.  $\therefore$  at a fixed volume, gas pressure results from particle collisions with the container walls, and at a fixed absolute temperature the collision frequency is directly proportional to the number of gas particles present. Therefore the measured pressure, volume, and temperature are related by  $PV = nRT$ , giving  $n = PV/RT$ .
197. 79N The piston is square, so its area is side<sup>2</sup>.  $\therefore$  pressure is force per unit area, the force equals the pressure multiplied by the piston area after converting all quantities to SI units.
198. 0.34.  $\therefore$  14.3 g of strontium carbonate corresponds to 0.0969 mol. The reaction produces 0.097 mol of carbon dioxide. The

amount of neon remains 0.19 mol. The mole fraction equals the moles of the requested gas divided by the total moles of gas present after the reaction.

199. 3.8 L.  $\therefore$  the amount of gas is conserved and the final pressure is atmospheric pressure, the gas expands until  $P_i V_i = P_f V_f$ . Because the final pressure is 1 atm, the final total volume is larger than the cylinder volume by the same factor as the initial pressure.
200.  $7.76 \times 10^5$  Pa.  $\therefore$  rearranging the combined gas law gives  $P_1 = P_2(V_2/V_1)(T_1/T_2)$ . Pressure is directly proportional to absolute temperature and inversely proportional to volume for a fixed amount of gas, so both changes must be accounted for simultaneously.